Pizza Sales SQL Query Analysis

A comprehensive guide to building SQL-driven KPI reports and daily trend charts for pizza restaurant operations. This document walks through essential queries for tracking revenue, orders, and product performance to help optimize business decisions.

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CORE METRICS

Essential KPIs for Restaurant Performance

Every successful pizza chain needs to track five fundamental metrics that tell the complete story of operational health. These KPIs provide instant visibility into revenue generation, customer behavior, and order patterns.

$801K

Total Revenue

Year-to-date sales performance

21K

Total Orders

Customer transactions processed

50K

Pizzas Sold

Total units moved

$38

Avg Order Value

Revenue per transaction

2.3

Pizzas Per Order

Average items purchased

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SQL QUERIES

Total Revenue Query

The revenue calculation aggregates all order amounts to provide the financial foundation for performance analysis. This query uses SUM aggregation on the total_price column to calculate gross revenue across all transactions.

Key considerations: Ensure your date filters align with fiscal periods, and account for any refunds or cancellations that might affect accuracy.

```
SELECT
 SUM(total_price) AS Total_Revenue
FROM pizza_sales;
```

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Average Order Value & Order Metrics

1

Average Order Value

Divides total revenue by number of distinct orders to determine the average transaction size. This metric helps identify pricing effectiveness and upsell opportunities.

```
SELECT
  SUM(total_price) / COUNT(DISTINCT order_id)
  AS Avg_Order_Value
FROM pizza_sales;
```

2

Total Orders Count

Counts distinct order IDs to measure transaction volume. Essential for understanding customer frequency and demand patterns throughout the business cycle.

```
SELECT
 COUNT(DISTINCT order_id) AS Total_Orders
FROM pizza_sales;
```

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Pizza Volume & Order Composition

Total Pizzas Sold

Aggregates the quantity column to calculate total pizza units sold across all orders. This volume metric is crucial for inventory planning and production scheduling.

```
SELECT
  SUM(quantity) AS Total_Pizzas_Sold
FROM pizza_sales;
```

Average Pizzas Per Order

Calculates the typical basket size by dividing total quantity by distinct orders. Helps optimize combo deals and portion sizing strategies.

```
SELECT
 CAST(CAST(SUM(quantity) AS DECIMAL(10,2)) /
 CAST(COUNT(DISTINCT order_id) AS
DECIMAL(10,2))
 AS DECIMAL(10,2)) AS Avg_Pizzas_Per_Order
FROM pizza_sales;
```

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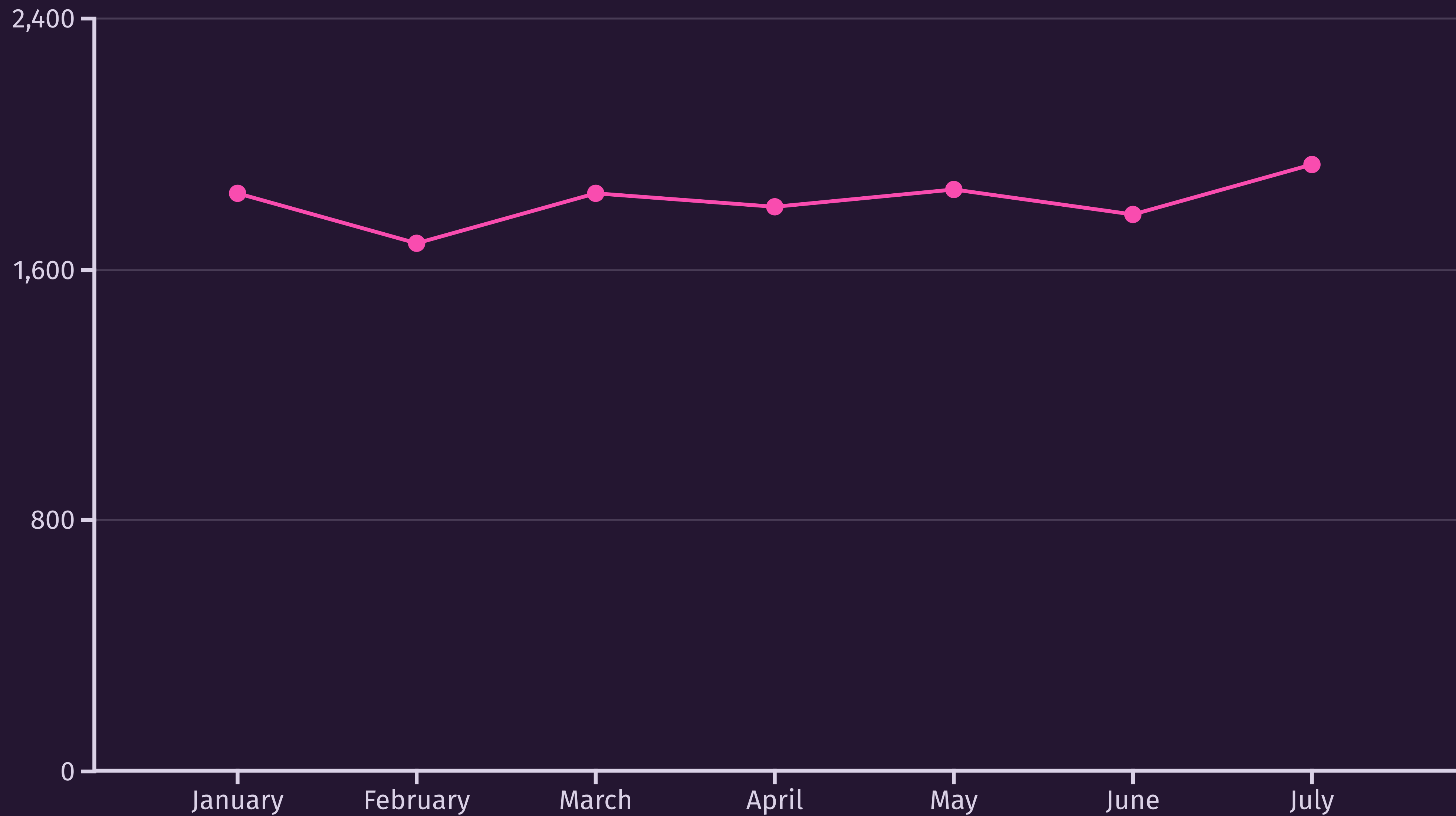
TREND ANALYSIS

Daily & Monthly Order Trends

Time-based analysis reveals patterns in customer behavior, helping optimize staffing and inventory. The daily trends show day-of-week patterns, while monthly trends identify seasonal fluctuations and growth trajectories.



Orders peak on Friday and Saturday evenings, while Sundays show strong afternoon demand. July represents the highest volume month, likely driven by summer events and holidays.

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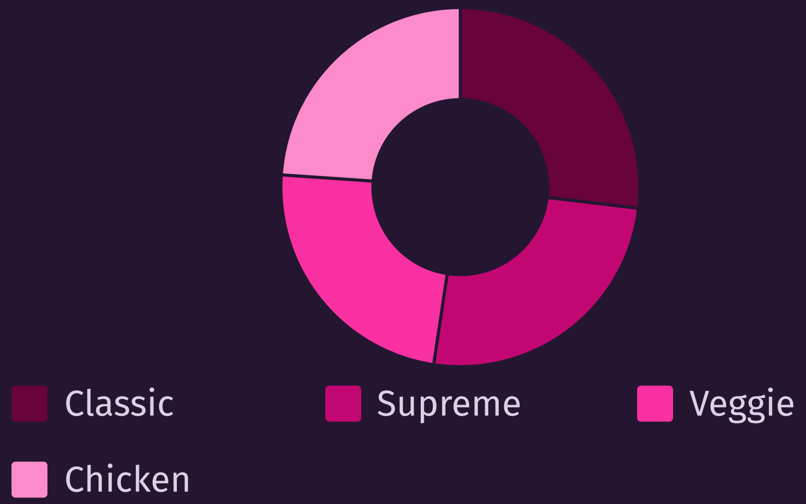
Sales Distribution by Category & Size

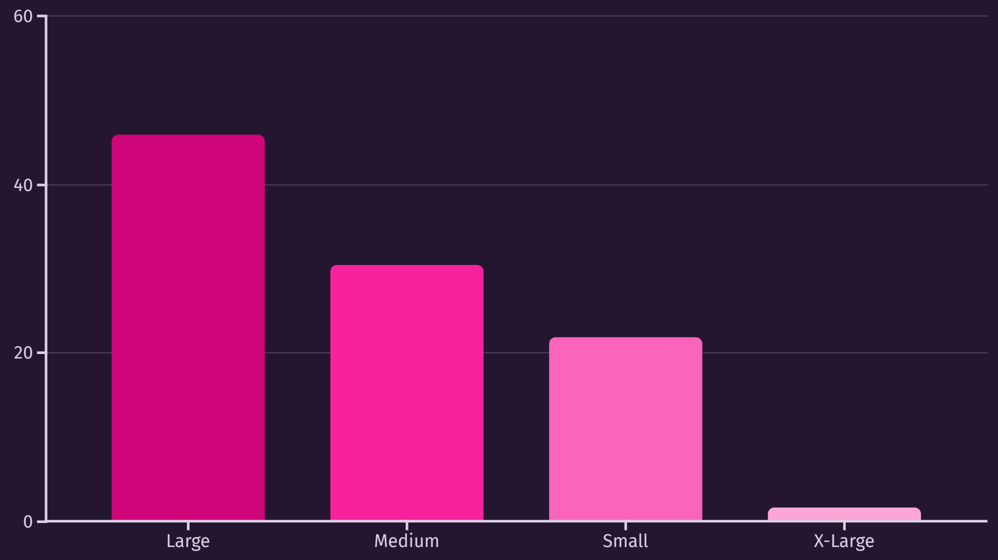
Category Performance

Breaking down sales by pizza category reveals which product lines drive revenue and should receive marketing focus.



Size Preference

Size distribution analysis helps optimize inventory and pricing tiers to match customer preferences.



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TOP PERFORMERS

Best-Selling Pizzas by Revenue

Identifying top performers helps prioritize menu placement, promotional efforts, and ingredient procurement. Use descending ORDER BY clauses to surface winners, and LIMIT to focus on the top 5.

1

The Thai Chicken Pizza

Leading revenue generator with $43,434 in sales. Exotic flavors appeal to adventurous customers seeking premium options.

2

The Barbecue Chicken Pizza

Strong second place at $42,768. Classic American flavor profile drives consistent reorders and family meal purchases.

3

The California Chicken Pizza

$41,410 in revenue. Fresh ingredients and health-conscious positioning attract millennial demographics.

4

The Classic Deluxe Pizza

Traditional favorite generating $38,181. Reliable choice for customers who prefer familiar combinations.

5

The Spicy Italian Pizza

$34,831 performance. Bold flavors and premium meats justify higher price points.

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PERFORMANCE ANALYSIS

Bottom 5 Pizzas Requiring Attention

Understanding underperformers is equally critical. Use ascending ORDER BY to identify pizzas that may need recipe improvements, better marketing, or potential menu removal. These insights drive strategic decisions about menu optimization.

The Brie Carre Pizza

Lowest revenue at $11,588 and only 490 units sold. Premium cheese may be too niche for mass market appeal.

The Mediterranean Pizza

$15,360 revenue with limited customer awareness. Consider repositioning as a health-focused option.

The Calabrese Pizza

$15,734 performance suggests regional flavor profile doesn't resonate with local customer base.

The Spinach Supreme Pizza

$15,277 revenue. Vegetable-forward options need stronger marketing to compete with meat-based pizzas.

The Soppressata Pizza

$15,642 sales indicate specialty meats require customer education and menu description improvements.

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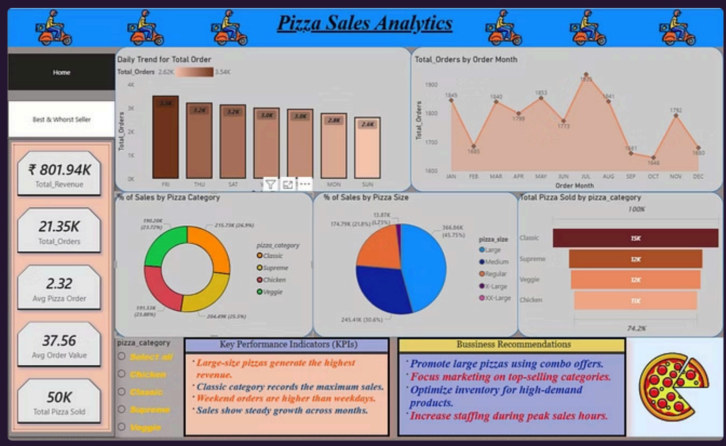

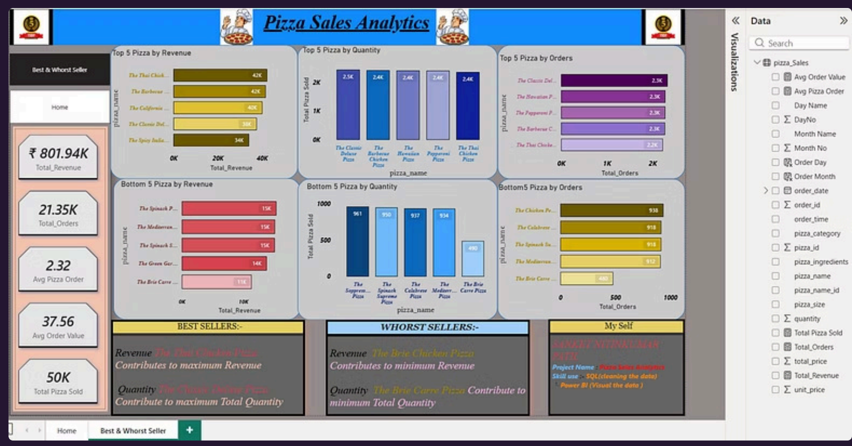

Please review and describe my Pizza Sales Analytics Dashboard project hosted on GitHub: Repository URL :- [Click here]

Sample Data

File upload
pizza_sales (1).csv
7.5 MB

Implementing KPIs in Power BI

Transform these SQL queries into actionable Power BI

Please review and describe my Pizza Sales Analytics Dashboard project hosted on GitHub: Repository URL :- [Click here]

I dashboards that update automatically. Connect your database, import queries as data sources, and build interactive visuals that enable drill-down analysis by date, category, and location.

01

Configure Data Source

Connect Power BI to your SQL database and import the pizza_sales table with refresh scheduling.

02

Create Calculated Measures

Build DAX formulas for KPIs like total revenue, average order value, and sales percentages.

03

Design Visual Hierarchy

Place key metrics at the top, followed by trend charts, category breakdowns, and top/bottom performers.

04

Add Interactive Filters

Enable date slicers, category filters, and location selectors for dynamic analysis capabilities.

05

Schedule Refresh & Share

Automate daily data updates and distribute dashboard access to stakeholders via Power BI Service.

> With these SQL queries and Power BI implementation, your team gains real-time visibility into pizza sales performance, enabling data-driven decisions that optimize menu offerings, staffing, and promotional strategies.

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About the Author & Project

Sanket N.Patil

Data Analytics & SQL Developer

Sanket is a passionate Data Analytics & SQL Developer with a proven track record of transforming raw data into actionable business intelligence. He specializes in leveraging SQL for robust data manipulation and Power BI for creating dynamic, insightful visualizations that drive informed decision-making.

Key Skills:

SQL, Power BI, Data Visualization, Database Management

Pizza Sales Analytics Dashboard

This project showcases a complete data analytics pipeline, from performing complex SQL queries to developing an interactive Power BI dashboard for in-depth pizza sales analysis.

Key Achievements:

- Analyzed over 21,000 orders to understand customer purchasing patterns.
- Tracked more than 50,000 individual pizza sales for comprehensive trend identification.
- Identified top and bottom performing pizzas, categories, and sales periods.

GitHub Repository: **Click here**

Technologies Used:

SQL Server, Power BI, Data Analysis

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